LIRA — Network Security Investigation Report

> **ML detects. LLM investigates, explains and guides.**

1. Executive Summary

- **Investigation ID:** `d25e09b9-1a95-45db-9ee7-ec1f44e0bc4a`
- **Status:** COMPLETED
- **Predicted traffic class:** `PortScan`
- **Random Forest confidence:** 100.0%
- **Additional analyses used:** 3
- **Analyses remaining:** 0
- **Report generated:** 2026-11-04 02:46 UTC

Preliminary assessment

The model classified this flow as **PortScan**. This is a model prediction, not independent proof that a real-world attack occurred.

2. Investigation History

- Reason: One analysis remains and dataset-label context has not been reviewed; need to clarify what PortScan label can and cannot prove before drawing provisional conclusion.

3. Investigation Evidence

Analysis 1 — `analyze_prediction`

- **Execution status:** SUCCESS
- **Timestamp (UTC):** 2026-11-04T02:46:43.470333+00:00
- **Reason:** Initial detection shows PortScan with 1.0 confidence but no evidence has been collected; need to assess prediction ambiguity and uncertainty before concluding.


```
{
  "predicted_class": "PortScan",
  "status": "ATTACK",
  "model_confidence": 1.0,
  "top_probability": 1.0,
  "second_best_class": "BENIGN",
  "probability_gap": 1.0,
  "prediction_uncertainty": "LOW",
  "requires_further_investigation": false,
  "interpretation": "The leading class is clearly ahead of alternatives.",
  "limitations": [
    "Random Forest probabilities may not be calibrated.",
    "A high probability does not prove an attack occurred."
  ]
}
```


Analysis 2 — `check_data_quality`

- **Execution status:** SUCCESS
- **Timestamp (UTC):** 2026-11-04T02:46:46.227479+00:00
- **Reason:** Model confidence is 1.0 but limitations note probabilities may be uncalibrated and do not prove an attack; need to verify input data quality and feature validity before drawing conclusions.


```
{
  "tool": "check_data_quality",
```

```
  "valid": true,
  "status": "VALID",
  "rows_checked": 1,
  "features_checked": 78,
  "missing_or_non_numeric_values": 0,
  "infinite_values": 0
}
```


Analysis 3 — `review_class_context`

- **Execution status:** SUCCESS
- **Timestamp (UTC):** 2026-11-04T02:46:48.096245+00:00
- **Reason:** One analysis remains and dataset-label context has not been reviewed; need to clarify what PortScan label can and cannot prove before drawing provisional conclusion.


```
{
  "tool": "review_class_context",
  "status": "COMPLETED",
  "predicted_class": "PortScan",
  "dataset_description": "Traffic associated with port-scanning activity in CIC-IDS2017.",
  "caution": "Scanning can have legitimate uses; investigate surrounding context.",
  "scope": "This describes dataset labels, not verified live incidents."
}
```


4. Final Assessment

Provisional conclusion: Model predicts PortScan (ATTACK) with 1.0 confidence and data quality is VALID, but this is a dataset-label prediction, not verified proof of a real-world attack. Limitations note probabilities may be uncalibrated and scanning can have legitimate uses. No corroborating network evidence was collected within budget, so a real PortScan cannot be confirmed and uncertainty remains high; further contextual/log investigation would be required to verify.

5. Recommended Operational Follow-up

1. Check relevant server, authentication, and network logs for corroborating evidence.
2. Verify whether the activity matches legitimate administrative or testing activity.
3. Review the alert with a network or security administrator before taking disruptive action.
4. Do not treat model confidence or similarity to reference flows as definitive proof of compromise.

6. Limitations

- The investigation concerns a dataset record, not live production traffic.
- Random Forest probabilities may not be calibrated.
- Similarity to reference flows does not independently confirm an attack.
- No independent operational logs are available to confirm compromise.

Generated by LIRA (Learning Intrusion Reasoning Agent). This report supports security triage and does not replace human verification.